Adnan Harun Dogan

Personal Information

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Employment

Mar 2025–Oct 2025
Graduate Intern, Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich
Supervisor: Prof. Dr. Christian Holz — *Zürich, Switzerland*

Feb 2023–Present
Research Assistant, Image Processing and Pattern Recognition Laboratory (ImageLab), METU
Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas — *Ankara, Turkey*

Oct 2021–Feb 2022
Undergraduate Intern, Center for Artificial Intelligence and Robotics (ROMER), METU
Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku — *Ankara, Turkey*

Jun 2021–Sep 2021
Undergraduate Intern, Cyber-Physical Systems Laboratory, University of California, Irvine
Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque — *Irvine, California, USA*

Education

2022–2025
Master of Science in Computer Engineering, Middle East Technical University (METU)
Advisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas — *Ankara, Turkey*
Thesis: Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation

2018–2022
Bachelor of Science in Computer Engineering, Middle East Technical University (METU)
Ankara, Turkey

Research Interests

My research centres on differentiable optimization for deep learning: how to propagate useful gradients through combinatorial and constrained optimization layers, and how to train networks end to end when the loss is ranking based or defined through optimal transport. I apply these ideas mainly to object detection. A second line of work is physics-informed and structured state estimation, in particular learned Kalman filters and conservation-aware neural surrogates for fluid dynamics, along with deep models for physiological signals such as EEG and inertial sensing.

Awards and Fellowships

2022–2023	TUBITAK 2247-C STAR'23 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2022	Best Paper Award by IEEE/SRMC 2022
2022	Graduated in the first quantile of the class by Middle East Technical University (METU)
2021–2022	TUBITAK 2247-C STAR'22 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2020–2021	TUBITAK 2247-C STAR'21 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2017	University Entrance Exam: ranked 1466th of 2.27M candidates by OSYM, Turkey

Publications

Peer-reviewed publications

9. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. ***International Conference on Robotics and Automation (IEEE ICRA)***. (arXiv, Code)
8. Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. ***IEEE Transactions on Visualization and Computer Graphics***. (DOI, Code)
7. Amin, M. A., **Dogan, A. H.**, Kuru, E. S., Sever, Y., & Angin, P. (2024). Misuse detection and response for orchestrated microservices based software. *International Conference on Advanced Information Networking and Applications (AINA 2024), Lecture Notes on Data Engineering and Communications Technologies, pp. 217-226.* (DOI, Data)
6. Yavuz, F., Cam, B. C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. ***European Conference on Computer Vision (ECCV 2024), Lecture Notes in Computer Science, vol 15117, pp. 93-109***. (DOI, arXiv, PDF, Code)
5. Sever, Y., & **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. *ITU Journal on Future and Evolving Technologies, 4(2), 383-388.* (DOI, PDF, Code)
4. Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, T. S., & Tuncel, B. M. (2023). Enhancing underground built heritage analysis with text mining: A case study on Cappadocia. *In M. Akkar Ercan & K. Aydiner (Eds.), Valorising Underground Built Heritage in Cappadocia (pp. 61-92). CNR Edizioni. ISBN 978-88-8080-605-9.* (ISBN, PDF)
3. Sever, Y., Ekinci, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., & Angin, P. (2022). An empirical analysis of IDS approaches in container security. *IEEE/SRMC'22, September 2022.* (DOI)
   - Best Paper Award
2. Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., & Karagoz, P. (2021). Explainability in irony detection. *Big Data Analytics and Knowledge Discovery (DaWaK 2021), Lecture Notes in Computer Science, vol 12925, pp. 152-157.* (DOI)
1. Demirel, B. U., **Dogan, A. H.**, & Al-Faruque, M. A. (2021). Two might do: A beat-by-beat classification of cardiac abnormalities using deep learning and domain-specific features. *Computing in Cardiology (CinC).* (DOI, PDF, Code)

Preprints

2. **Dogan, A. H.**, Deniz, M., Alemdar, H., & Baran, O. U. (2026). *CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws.* arXiv preprint arXiv:2607.06587. (arXiv)
1. **Dogan, A. H.**, & Dogan, A. (2021). *An assembled deep learning approach for flow field prediction.* (PDF)

Manuscripts under review

3. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). *Deep Kalman state estimation for drift-calibrated full-body pose tracking from sparse inertial sensors.* Under Review at Conference on Computer Vision and Pattern Recognition.
2. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). *Magnetometer-free state estimation for sparse inertial human pose tracking via motion-based yaw inference.* Under Review at Conference on Computer Vision and Pattern Recognition.

1. **Dogan, A. H.**, Celtik, S. O., Baran, O. U., & Alemdar, H. (2026). *Prediction-driven adaptive mesh refinement with graph neural networks for transonic airfoil flows.* Under Review at Journal of Aerospace Science and Technology.

Theses

1. **Dogan, A. H.** (2025). *Incorporating piecewise linear functions with constant regions in backpropagation.* M.Sc. Thesis, Middle East Technical University. (Thesis)

Research Experience

Sep 2021–Jun 2022

Cyber Security Research Group, METU
Undergraduate Research Project
Advanced research on container security with intrusion detection systems (CONTSEC) for cloud-native environments. Designed and implemented high-performance IDS pipeline optimized for containerized cloud infrastructures. Collaborated on Kubernetes misuse detection research, contributing to novel datasets and detection methodologies.
Advisor: Assoc. Prof. Dr. Pelin Angin — *Ankara, Turkey*

Aug 2021–Oct 2021

TUBITAK BILGEM
Summer Intern
Cloud infrastructure development at Cloud Computing and Big Data Research Lab. Contributed to Sapphire Cloud platform enhancements.
Kocaeli, Turkey

Oct 2020–Oct 2021

Textual Event Graph HUB (TEGHub), METU
Undergraduate Research Project
Textual analysis for irony detection and research impact prediction. Implemented end-to-end neural pipelines integrating GCN and LSTM networks for analyzing citation patterns and textual content on Twitter datasets.
Advisor: Prof. Dr. Pinar Karagoz — *Ankara, Turkey*

Aug 2020–Oct 2020

Enocta Inc.
Summer Intern
NLP and transformer model optimization. Engineered GPT-2 and T5 models using HuggingFace for enterprise learning management systems.
Ankara, Turkey

Feb 2020–Jun 2021

School of Medicine, Koç University
Undergraduate Research Project
Healthcare policy simulation for evidence-based decision making. Utilized simulation models focusing on mechatronics applications in healthcare. Developed GCN-based citation graph embeddings for healthcare policy papers.
Advisor: Assist. Prof. Dr. Ozge Karanfil — *Istanbul, Turkey*

Jan 2020–Jun 2022

Image Processing and Pattern Recognition Laboratory (ImageLab), METU
Undergraduate Research Project
CNNFOIL: Approximating HLLC Riemann solver for flow prediction around airfoils via encoder-decoder neural networks. Engineered state-of-the-art neural network models achieving 4.4% performance improvement in predicting critical flow field parameters around airfoil geometries.
Advisor: Assist. Prof. Dr. Hande Alemdar, Assist. Prof. Dr. Baran Ugras — *Ankara, Turkey*

Teaching

Teaching Assistant, Computer Engineering Department, METU Feb 2022–Present

8. CENG 280 - Automata Theory (Spring 2024, Spring 2025, Spring 2026)
7. CENG 242 - Programming Language Concepts (Spring 2022, Spring 2023, Spring 2025, Spring 2026)
6. CENG 223 - Discrete Computational Structures (Fall 2023, Fall 2024, Fall 2025)
5. CENG 424 - Logic for Computer Science (Fall 2023, Fall 2025)
4. CENG 460 - Introduction to Robotics for Computer Science (Fall 2024)
3. CENG 213 - Data Structures (Spring 2024)
2. CENG 382 - Analysis of Dynamical Systems (Spring 2023)
1. CENG 334 - Introduction to Operating Systems (Spring 2022)

Undergraduate Teaching Assistant, Computer Engineering Department, METU Spring 2019–Fall 2021

3. CENG 240 - Programming with Python for Engineers (Fall 2021)
2. CENG 111 - Introduction to Computer Engineering Concepts (Fall 2020)
1. CENG 230 - Introduction to C Programming (Spring 2019)

Computer Skills

Languages & Tools:	Python/PyTorch, C/C++, ROS2, Rust
DevOps & HPC:	GitHub, Docker, Slurm (Workload Manager), Apptainer
Differentiable Optimization:	Combinatorial Optimization, Optimal Transport, Deep State Space Models
Deep Learning:	Computer Vision, Self-Supervised Learning, Probabilistic Deep Learning, Deep Reinforcement Learning

Languages

Turkish:	Native
English:	Fluent (TOEFL iBT 109/120, 2022)